**THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA**

**58th NMTC - SCREENING TEST - GAUSS CONTEST**

**PRIMARY LEVEL - V & VI GRADES**

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**PART – A**

**Note :** Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.

**1.** The three-digit number 3a4 is added to 278 to obtain another three-digit number 6b2, here a and b are digits. If 6b2 is divisible by 9, then a + b =

(A) 2 (B) 4 (C) 6 (D) 7

**2.** In an AMTI workshop, the ratio of number of boys to the number of girls was 5:8. If there are total 65 students in the workshop, how many more girls were there in the workshop than boys?

(A) 40 (B) 15 (C) 26 (D) 30

**3.** A person covers first 1.5 km at 5 km/h and next 3 km at 6 km/h, then his average speed is

(A) $\frac{45}{16}$ km / h (B) $\frac{45}{8}$ km / h (C) 4 km/h (D) 5.5 km/h

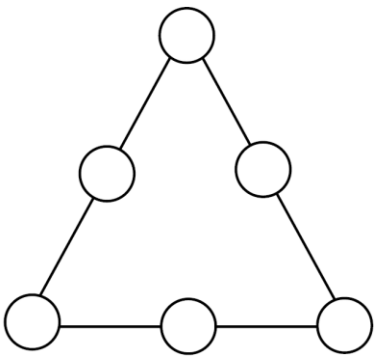
**4.** A triangle and a square have equal perimeters. The lengths of three sides of the triangle are 7.4 cm, 5.7 cm, 6.9 cm. Then area of the square is _____ square cm.

(A) 100 (B) 81 (C) 36 (D) 25

**5.** Ramanujan Primary School has 1200 students. Each class has 30 students. Each student takes 5 sessions a day and each teacher teaches 4 sessions a day. What is the minimum number of teachers in the School?

(A) 40 (B) 100 (C) 50 (D) 20

**6.** In a magic triangle of addition, each of six natural numbers from 21 to 26 is placed in one of the six empty circles such that the sum S of the three numbers from the circles along each side would be the same. Then the maximum possible value of S is



(A) 84 (B) 80 (C) 72 (D) 69

**7.** A ream of identical papers containing 500 sheets is 5 cm thick. How many sheets of this type of paper would be there in a 7.2 cm thick bundle?

(A) 772 (B) 720 (C) 750 (D) 680

**8.** n = 1111…11 is a 2026-digit number with each digit as 1. n is divided by 1111, then the quotient is Q and the remainder is R. then

(A) Q is a 2026-digit number and R = 11 (B) Q is a 2023-digit number and R = 11

(C) Q is a 1517-digit number and R = 11 (D) Q is a 2023-digit number and R = 111

**9.** A 'stair-step' figure is made of alternating black and white squares in each row. All rows begin and end with white square. First four figures of the stair-step figure are shown below. Fig-1 has 1 white square, Fig-2 has 3 white and 1 black squares, Fig-3 has 6 white and 3 black squares, … and it continues. What is the the total number of white squares in the Fig-20?

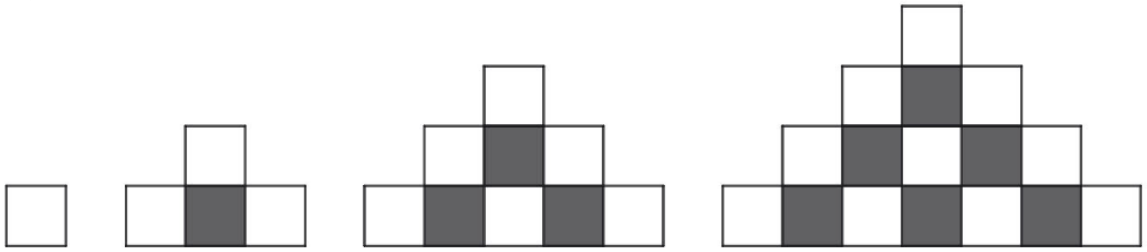


Fig. 1 Fig. 2 Fig. 3 Fig. 4

(A) 220 (B) 210 (C) 200 (D) 190

**10.** m, n, p are all different natural numbers each between 2 and 9. $\frac{m+n+p}{m+n}$ is an integer, then the number of all possible value of $\frac{m+n+p}{m+n}$ is

(A) 1 (B) 2 (C) 3 (D) 8

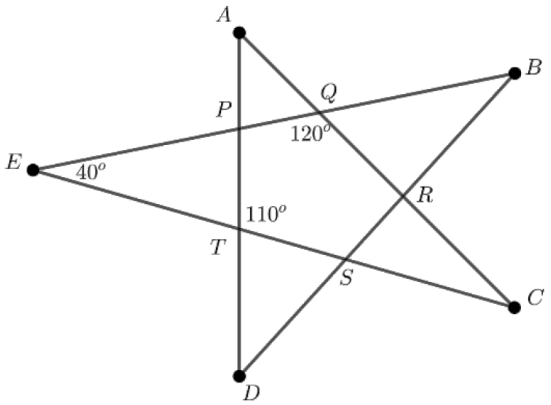
## PART - B

**Note :** Write the correct answer in the space provided in the response sheet.

**11.** $1\frac{7}{20} \div 2.7 + 2.7 \div 1.35 + \left(0.4 \div 2\frac{1}{2}\right) \times \left(4.2 - 1\frac{3}{40}\right) =$ _____

**12.** a and b are natural numbers such that a < b and a × b = 2002. Then the minimum value of b – a is _____.

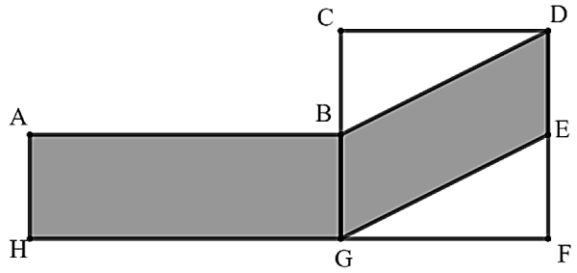
**13.** In the adjoining figure, measure of ∠PQR = 120°, measure of ∠PET = 40°, and measure of ∠PTS = 110°, then the measure of $\angle PAQ$ = _____.




**14.** a, b, c, d, e are five consecutive natural numbers. Another natural number n always divides (a × b × c × d × e) completely. Then the greatest possible value of n is _____.

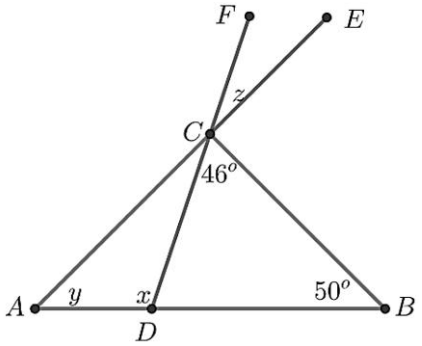
**15.** An army general wanted to make a square arrangement of his soldiers. There were 2350 soldiers. After completing the square arrangement, he found that 46 soldiers were left over. Hence the number of soldiers in each row of the square arrangement is _____.

**16.** ABGH is a rectangle, CDFG is a square. B is the midpoint of CG and E is the midpoint of DF. The length of the rectangle ABGH is three times of its breadth. The area of the shaded region is 80 $cm^2$. Then the perimeter of ABCDFGHA is _____ cm.



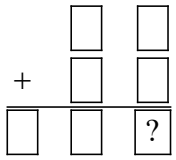

**17.** 80% of a is equal to 16% of b. b is x% of a, then 20% of x is _____.

**18.** In the adjoining figure, ΔABC is isosceles triangle with CA = CB. ∠ABC = 50°, point D is on side AB such that ∠BCD = 46°. Let ∠CAB = y, ∠ADC = x, ∠FCE = z, then x – y + 2z = _____°.



**19.** A man drove from 7:45am to 9:15am (on the same day) and covered 84 kilometer. Then his driving speed is _____ km/h.

**20.** In the addition of two 2-digit numbers, each blank space, including those in the answer, is to be filled with one of the digits 0, 1, 2, 3, 4, 5, 6, each used exactly once. The unit's place digit of the sum is _____.




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